

EngageAI PRD v1.1 - Recommended Fixes

1. Replace PostgreSQL with MySQL (XAMPP)

Replace every reference to PostgreSQL/Postgres with MySQL (XAMPP, local development). Update architecture, technology stack, database schema descriptions, MVP scope, build checklist, and Round 1 submission content accordingly.

2. Remove ChromaDB / Pinecone / RAG

Remove all references to ChromaDB, Pinecone, vector databases, embeddings, embedding_id fields, semantic retrieval, and RAG. Opportunity Discovery should read directly from the product_catalog table in MySQL and let the LLM rank products.

3. Update Technology Stack

Frontend: React + TypeScript + Tailwind CSS + shadcn/ui. Backend: FastAPI. Database: MySQL (XAMPP). Agent Orchestration: LangGraph. LLM Providers: GroqCloud (primary), Cerebras (synthetic customer generation), Gemini (fallback). Remove OpenAI and AWS references.

4. Remove AWS

Remove AWS deployment and Infrastructure-as-Code references. State that the MVP runs locally for the hackathon demo.

5. Simplify LangGraph for MVP

Implement a sequential workflow: Behavior Analyzer → Life Event Detection → Opportunity Discovery → Engagement Strategy → Action Planning → Follow-Up. Advanced routing can be deferred.

6. Add chat_messages table

Add table: chat_messages(id, customer_id, role, message, created_at) for conversation persistence.

7. Customer Timeline UI

Add a timeline component showing salary credits, detected life events, recommendations, accepted actions, and follow-ups.

8. Confidence Scores

Display confidence values for detected life events (e.g., Salary Hike – 91%).

9. Agentic AI Explanation

Add a short section comparing a traditional chatbot (Ask → Answer) with EngageAI (Observe → Analyze → Detect → Reason → Recommend → Act → Follow-up → Observe Again).

10. Freeze PRD

After these updates, freeze the PRD and focus entirely on implementation quality, UI polish, and the live demo.